

ADVANCED MACHINE LEARNING · WORKSHEET 09

Bias, Variance and the Bias–Variance Tradeoff

Estimators · Underfitting & Overfitting · MSE Decomposition · Practical Diagnosis

TEACHER COPY — Lecture 9

PART I: UNDERSTANDING BIAS AND VARIANCE

SECTION 1 Setting the Stage: $Y = f(x) + \varepsilon$

▷ HOOK

Suppose we want to model a student's **Stipend** which depends on three input features: CGPA (X_1), IQ (X_2), and Number of Projects (X_3).

There exists a **true hidden relationship** or function that determines the Stipend. But even then, the observed value will not be exactly equal to the true pattern, because there is always some randomness in the system.

We represent this randomness with random noise (ε):

$$\text{Stipend} = f(\text{CGPA}, \text{IQ}, \text{Projects}) + \varepsilon$$

Let $X_1 = \text{CGPA}$, $X_2 = \text{IQ}$, $X_3 = \text{Projects}$, $Y = \text{Stipend}$.

$$\Rightarrow Y = f(X_1, X_2, X_3) + \varepsilon$$

Let us just call the combination of input variables X_1, X_2, X_3 simply as X :

$$Y = f(X) + \varepsilon$$

where:

- $f(X) = \frac{\text{Actual hidden relationship}}{\text{True function}}$
- $Y = \frac{\text{Observed Output}}{\text{Target Variable}}$
- $\varepsilon = \frac{\text{Random Noise}}{\text{Irreducible Error}}$

In words:

$$\text{Observed Output} = \text{True Pattern} + \text{Random Noise}$$

Properties of the noise:

By definition, the mean of all random noise (white noise) is 0:

$$E(\varepsilon) = \underline{0}$$

Say ε has some variance σ^2 , where σ is the standard deviation of white noise:

$$\text{Var}(\varepsilon) = \sigma^2 = E(\varepsilon^2) - [E(\varepsilon)]^2 = E(\varepsilon^2) - 0$$

$$\Rightarrow \text{Var}(\varepsilon) = \sigma^2 = \underline{E(\varepsilon^2) - [E(\varepsilon)]^2}$$

What does white noise look like?

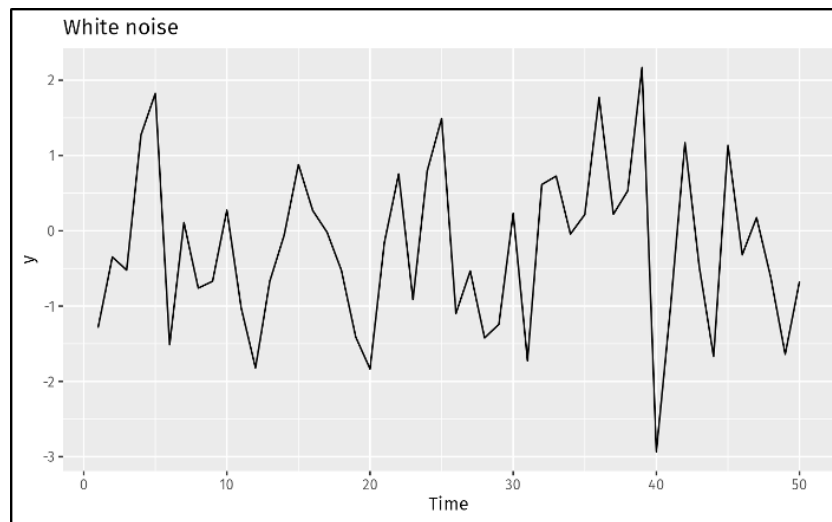


Fig. 1: White noise — random fluctuations with zero mean.

SECTION 2 The Estimator and the Random Variable

▷ HOOK

The true function $f(X)$ exists for the whole population of data. We don't know what this true function is. We can try to best estimate it using our **sample dataset** (which is derived from the population).

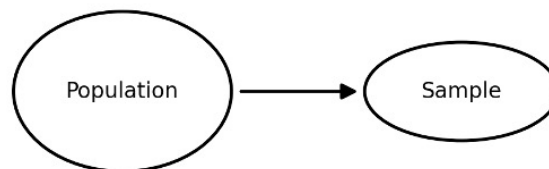


Fig. 2: We draw a sample from the larger population.

Using the training dataset, we build a model: $\hat{f}(x)$. This is called an **estimator** of the true function.

Examples of estimators:

1. Sample mean estimates sample population mean
2. Linear Regression estimates relationship
3. Decision tree estimates mapping from inputs to outputs

Imagine collecting **different training datasets** from the same population:

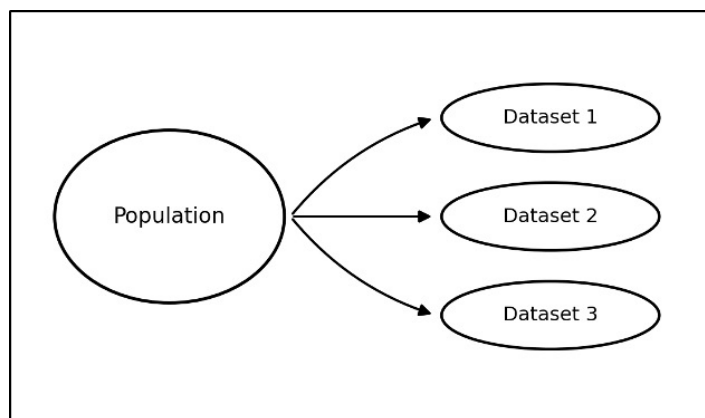


Fig. 3: Different sample datasets drawn from the same population.

★ KEY INSIGHT

- Sample Dataset 1 $\rightarrow$ Model 1
- Sample Dataset 2 $\rightarrow$ Slightly Different Model 2
- Sample Dataset 3 $\rightarrow$ Slightly Different Model 3

If the training sample data changes, the estimator also changes. So, $\hat{f}(x)$ is **random**. Different sample datasets produce different learned models.

So mathematically: $\hat{f}(x)$ is a random variable for a fixed input x .

In the equation $Y = f(x) + \varepsilon$, even for the same input X , the value of Y can be different because of random noise ε . Thus, Y can also be treated as a random variable.

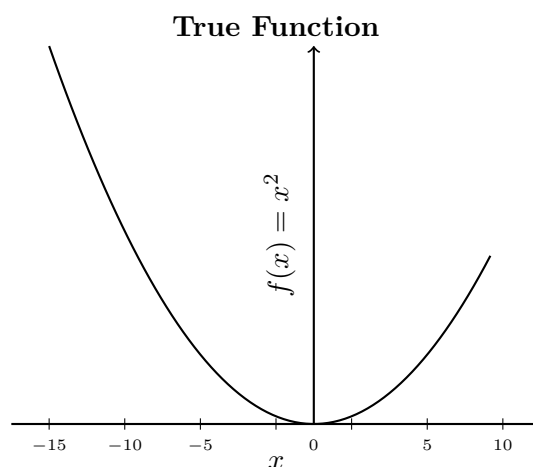
SECTION 3 Scenario: The God's-Eye View

▷ HOOK

Now consider yourself as a “God” who knows the true function of a population:

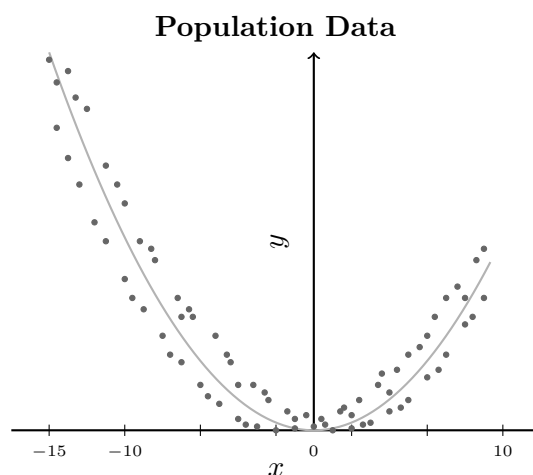
$$Y = f(x) = x^2 \quad \text{where } x \in [-15, 10]$$

Visual: True function $f(x) = x^2$



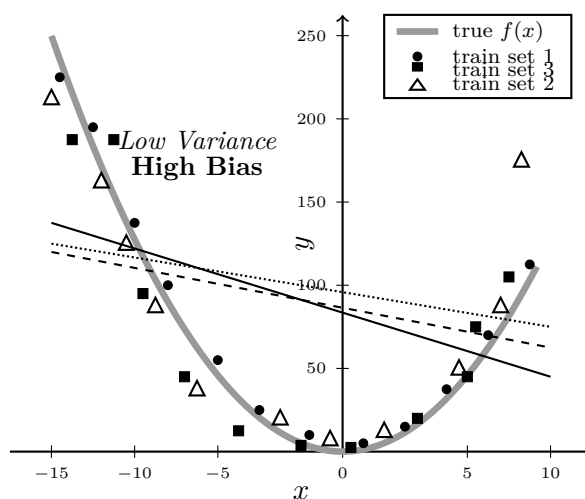
Now suppose we add random noise to this function to get 100 new data points. These become the **population data**.

Visual: Population data (true function + noise)



Now suppose we draw out **3 random sample sets** from this population data and give each to a different person. Each person builds an ML model.

3.1 Case 1: All three build a straight-line model (without knowing the true function)



All 3 datasets and their straight-line models are shown on a single graph. Notice that the three lines are very close to each other (low variance) but none of them can capture the parabolic shape (high bias).

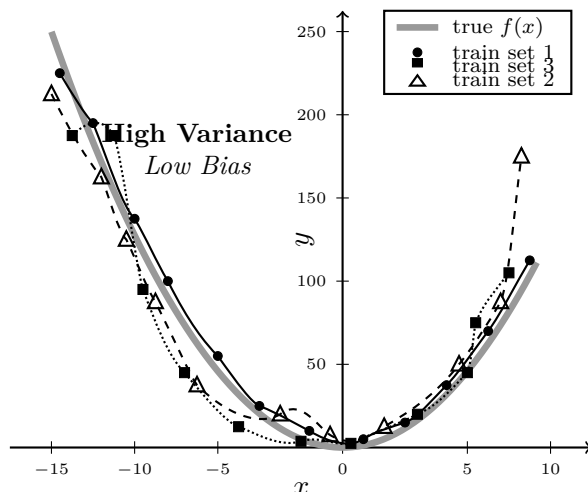
All three linear models are **not able to fit** their respective training data properly.

These models have **High Bias**: They are too simple to capture the curve

These models have **Low Variance**: The 3 lines are very similar to each other

These 3 models are underfitted models.

3.2 Case 2: All three increase the degree of their models



All 3 datasets and their high-degree models are shown on a single graph. Each curve passes through its own training points (low bias) but the three curves are wildly different from each other (high variance).

These models are trying to **exactly fit** their training datasets.

These models have **Low Bias**: They fit their specific training data extremely well

These models have **High Variance**: The 3 curves are very different from each other

These 3 models are overfitted models.

PRACTICE P1 · BIAS AND VARIANCE INTUITION

(a) Complete the table:

Model type	Bias	Variance	Fitting behaviour
Too simple (e.g. straight line for x^2)	<u>High</u>	<u>Low</u>	<u>Underfits</u>
Too complex (e.g. degree-20 polynomial)	<u>Low</u>	<u>High</u>	<u>Overfits</u>
Appropriately complex	<u>Low</u>	<u>Low</u>	<u>Good fit</u>

(b) **True or False:** If the training data changes, an underfitting model's predictions will change drastically. **[F]**

Justify: An underfitting model has low variance. Its predictions are mostly stable regardless of training data changes.

(c) **True or False:** A model with high bias always has high variance. **[F]**

Justify: They are generally inversely related. High bias models are too simple and thus have low variance.

SECTION 4 The Bias–Variance Tradeoff

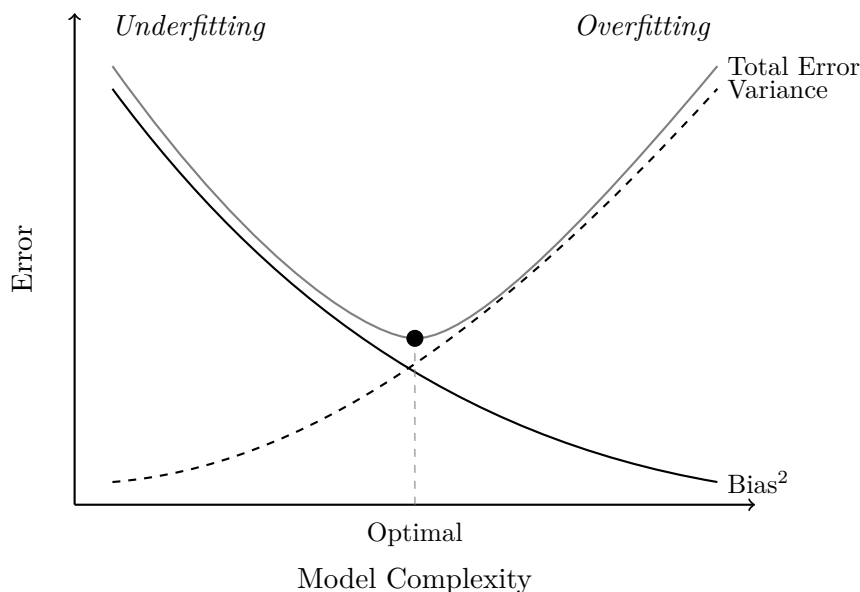
▷ HOOK

When you try to **decrease the bias** of a model, its variance automatically starts **increasing**. This is known as the **Bias–Variance Tradeoff**.

When Bias $\uparrow$, Variance $\downarrow$ and vice versa.

Both Bias and Variance are forms of errors. Their relationship with model complexity follows this graph:

Visual: Bias–Variance Tradeoff



We aim to reach the model complexity where the **total error is minimum**.

SECTION 5 Revisiting the Discussion: Mathematical Definitions

▷ HOOK

Let us revisit the equation:

$$Y = f(x) + \varepsilon$$

where $f(x)$ = true population function, Y = observed output, ε = random noise = irreducible error.

$$E(\varepsilon) = 0, \quad \text{Var}(\varepsilon) = \sigma^2 = E(\varepsilon^2).$$

We have an estimator of the true function, $\hat{f}(x)$, which is also a random variable.

5.1 Bias

$$\text{Bias}(\hat{f}(x)) = E[\hat{f}(x)] - f(x)$$

where:

- $E[\hat{f}(x)]$ = Expected value of the prediction over different sample training datasets
- $f(x)$ = True function

Unbiased Predictor:

$$\text{Bias}(\hat{f}(x)) = 0 \implies E[\hat{f}(x)] = \underline{\mathbf{f(x)}} \text{ for all } x.$$

Example: Say you have 5 models with slopes m_1, m_2, m_3, m_4, m_5 . If $E(m) = \gamma$ or $\frac{\sum m_i}{5} = \gamma$, where γ is the slope of the true function, then we can say the model is an unbiased estimator.

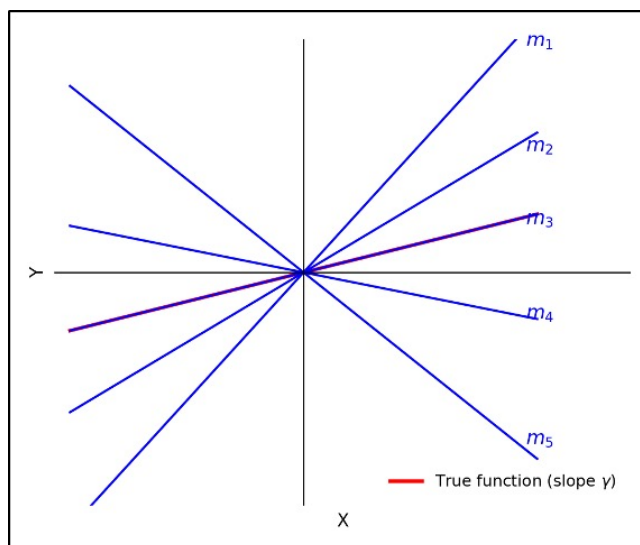


Fig. 4: Five models with different slopes (m_1 – m_5) and the true function (slope γ).

5.2 Variance

Mathematically, we calculate the variance of our predictor:

$$\text{Var}(\hat{f}(x)) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

Interpretation:

- Train on many sample datasets.
- Observe prediction spread.

SECTION 6 MSE and the Bias–Variance Decomposition

6.1 Mean Squared Error (MSE)

At a point x :

$$\text{MSE}(x) = E[(Y - \hat{f}(x))^2]$$

This measures the expected prediction error.

6.2 Bias–Variance Decomposition — Final Result

$$\text{MSE} = \text{Bias}^2 + \text{Variance} + \text{Var}(\varepsilon)$$

or equivalently:

$$\text{MSE} = \underbrace{\text{Bias}^2 + \text{Variance}}_{\text{Reducible Error}} + \underbrace{\text{Var}(\varepsilon)}_{\text{Irreducible Error}}$$

⚠ WARNING

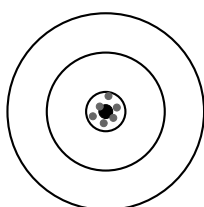
The proof of the above Bias–Variance Decomposition result is beyond the scope of this lecture. Students who are interested in the proof are welcome to ask the instructor about it separately after the lecture.

SECTION 7 Graphical Illustration of Bias and Variance

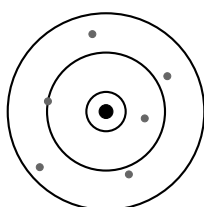
Visual: Dartboard analogy

Imagine throwing darts at a dartboard multiple times. Each throw represents training a model on a different sample dataset and making a prediction.

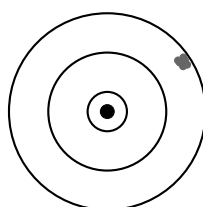
- **Concentric circles** represent increasing distance from the true value — the further out, the larger the error.
- **Bold dot at the centre** (bull's-eye) represents the **true value** we are trying to predict.
- **Smaller dots** represent the **predictions** obtained from models trained on different sample datasets.



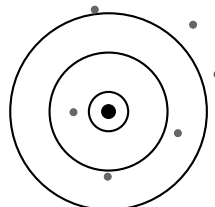
Low Bias,
Low Variance



Low Bias,
High Variance



High Bias,
Low Variance



High Bias,
High Variance

PRACTICE P2 · MATHEMATICAL CONCEPTS

- (a) If $E[\hat{f}(x)] = 12$ and $f(x) = 10$ at a particular point x , what is the bias?

Bias = $\underline{12 - 10 = 2}$

- (b) If the bias is 2, the variance is 3, and $\sigma^2 = 1$, what is the MSE?

MSE = $\underline{2^2 + 3 + 1 = 8}$

- (c) **Match the Column.** Match each dartboard pattern with the correct characterisation.

	Dartboard Pattern	Characterisation
1	Dots tightly clustered around the bullseye	(A) High Bias, High Variance
2	Dots tightly clustered far from the bullseye	(B) Low Bias, Low Variance
3	Dots spread widely around the bullseye	(C) High Bias, Low Variance
4	Dots spread widely far from the bullseye	(D) Low Bias, High Variance

Answers: 1 → B 2 → C 3 → D 4 → A

- (d) **True or False:** Irreducible error can be reduced by choosing a more complex model. **[F]**

Justify: Irreducible error is inherent noise in the system, it cannot be reduced.

- (e) A model has Bias² = 1, Variance = 9, σ^2 = 2. What is the total error? What component dominates?

Total error = 1 + 9 + 2 = 12. The Variance component dominates.

PART II: BRIDGING THEORY WITH PRACTICE

SECTION 8 The Mathematical World vs. The Real World

▷ HOOK

All this mathematics makes sense. But in real life, how do we actually calculate Bias and Variance? This is perhaps one of the biggest differences between learning Machine Learning **theoretically** and doing Machine Learning **in practice**.

Question 1: In real-world problems, do we have access to the true function $f(x)$? Why or why not?

No. If we already knew the true function, we would never need Machine Learning!

Question 2: When you download a dataset (e.g. from Kaggle), does it represent the entire population? Explain.

No. That dataset is usually just one sample collected from a much larger, unknown population

So mathematically, we can think of it like this:

Unknown Population $\xrightarrow{\text{sample}}$ One Sample Dataset (e.g. Kaggle CSV) $\xrightarrow{\text{train}}$ Train One ML Model

★ KEY INSIGHT

We only have **one sample dataset**, not hundreds of different sample datasets like we imagined while defining Bias and Variance.

Therefore, the true mathematical Bias and Variance **cannot be calculated** in real-world ML problems. Their definitions require:

- The true function $f(x)$ — which is unknown.
- Predictions from many models trained on many different sample datasets — which we do not have.

The mathematical definitions of Bias and Variance are **theoretical definitions**. They help us understand how learning algorithms behave, but they usually cannot be computed exactly in practice.

Then why do we study them?

Because they explain:

- why models make errors
- why overfitting happens
- why underfitting happens
- why increasing model complexity is not always beneficial

Even though we cannot compute the exact values, the **concepts guide almost every important decision** in ML.

SECTION 9 Diagnosing High Bias and High Variance

▷ HOOK

Instead of trying to calculate the exact Bias and Variance, practitioners try to **infer** whether a model is suffering from high Bias or high Variance.

Think of a doctor. A doctor usually cannot directly measure pain. Instead, the doctor looks at **symptoms**. Similarly, an ML engineer looks at evaluation results to **diagnose** the model.

9.1 Diagnosing High Bias (Underfitting)

Training Accuracy = 70%

Validation Accuracy = 68%

Both are low. If the model cannot even learn the training data properly, it is probably too simple.
This is a sign of **High Bias (Underfitting)**.

9.2 Diagnosing High Variance (Overfitting)

Training Accuracy = 99%

Validation Accuracy = 82%

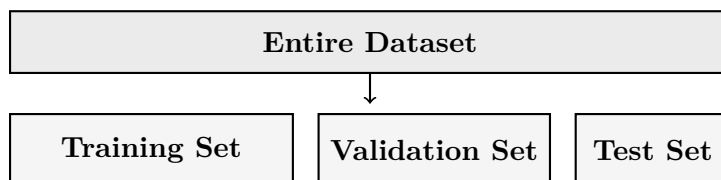
The model performs extremely well on training data but much worse on unseen data. It has **memorized** the training data instead of learning general patterns.

This is a sign of **High Variance (Overfitting)**.

SECTION 10 Why Do We Split the Dataset?

▷ HOOK

In practice, we cannot collect many different datasets from the population. Instead, we try to **simulate the idea** by splitting our available dataset.



- The **training set** is used to learn the model.
- The **validation set** is used to check whether the model generalizes to unseen data.

Diagnosing from splits:

- Model performs well only on training data, poorly on validation data $\Rightarrow$ suspect high variance.
- Model performs poorly on both $\Rightarrow$ suspect high bias.

Although this does not calculate Bias and Variance mathematically, it gives us **practical evidence** about the behaviour of the model.

SECTION 11 Cross-Validation

Suppose we split the dataset **only once**. The result may depend heavily on that particular split.

To reduce this dependence, we use **Cross-Validation**. Instead of training the model once, we train it **multiple times** using different train-validation splits. This allows us to evaluate how stable the model is across different subsets of the available data.

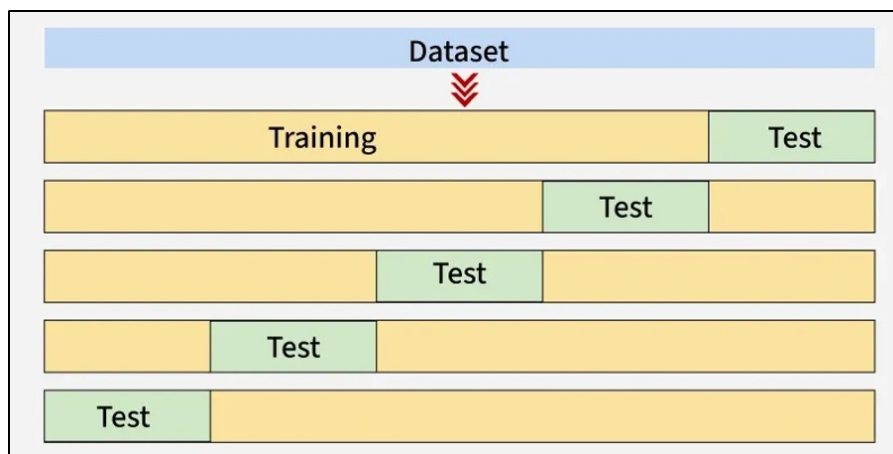


Fig. 5: K-Fold Cross-Validation — each fold takes a turn as the test set.

If the performance changes significantly across different folds, it suggests that the model is **sensitive to the training data** — a practical indication of higher variance.

◆ TAKEAWAY

You can study Cross-Validation in detail as **homework**. The key idea is: training the model multiple times on different splits helps us estimate how much the model's performance varies — which is closely related to the concept of Variance.

SECTION 12 Learning Curves Give Another Clue

▷ HOOK

A **learning curve** shows how the training and validation errors change as we increase the amount of training data. Without ever knowing the true function, these curves help us understand what kind of problem our model is facing.

Visual: Training Error and Validation Error vs. Model Complexity

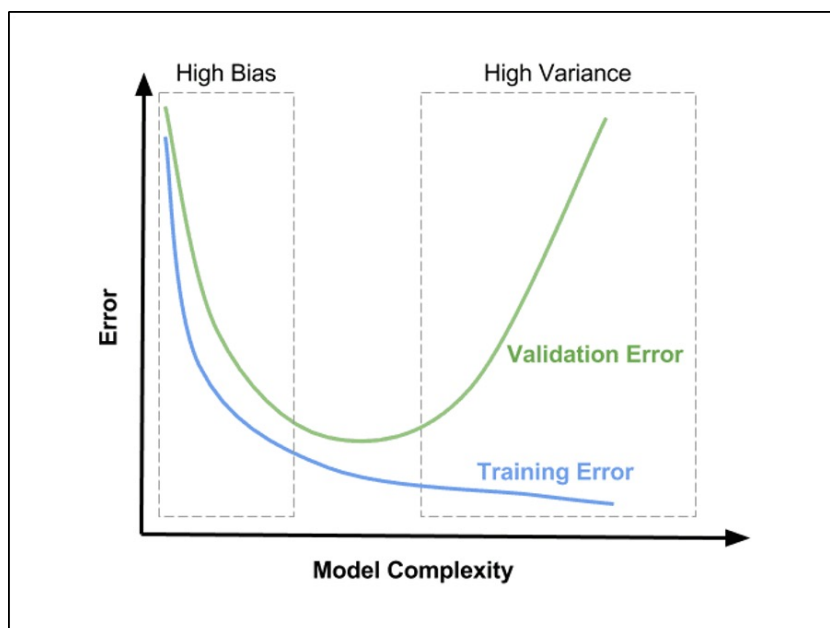


Fig. 6: How training and validation errors behave as model complexity increases.

Typical observations:

- **High Bias:** Both training and validation errors remain high, even after adding more data.
- **High Variance:** Training error is very low, but validation error is much higher. As more data is added, the validation error often decreases.

PRACTICE P3 · DIAGNOSIS AND APPLICATION

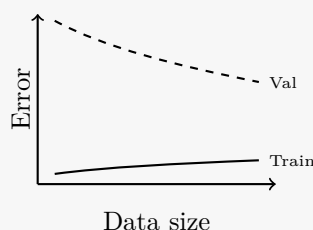
- (a) For each scenario, diagnose whether the model suffers from **High Bias**, **High Variance**, or a **Good fit**.

Scenario	Train Acc.	Val Acc.	Diagnosis
(i)	98%	97%	<u>Good fit</u>
(ii)	65%	63%	<u>High Bias</u>
(iii)	99%	72%	<u>High Variance</u>
(iv)	55%	54%	<u>High Bias</u>

- (b) A team trains a decision tree with no depth limit on a house price dataset. Training RMSE is very low but validation RMSE is 3 times higher.

- (i) What problem is this? High Variance (Overfitting)
- (ii) Name **two** ways to fix it: Limit tree depth (pruning) and Add more training data

- (c) Look at the learning curve below and answer:



- (i) Is this High Bias or High Variance? High Variance
- (ii) Will adding more training data likely help? Yes (Val curve goes down)
- (iii) Would increasing model complexity help? No, it would make it worse

- (d) **Assertion–Reason:**

A: A linear regression model applied to a highly non-linear dataset is likely to have high bias.

R: A linear model cannot capture non-linear patterns, so it will systematically underfit the data.

- (i) Both A and R are true, and R is the correct explanation of A. **[T]**
- (ii) Both A and R are true, but R is not the correct explanation of A. **[]**
- (iii) A is true but R is false. **[]**
- (iv) A is false but R is true. **[]**

SECTION 13 Final Takeaway**▷ HOOK**

Let us summarise everything we have learned.

Mathematically:

- Bias and Variance are defined over all possible sample training datasets drawn from the same population.
- Their definitions assume that the true function $f(x)$ is known.
- Therefore, they are **theoretical quantities** that help us understand how learning algorithms behave.

Practically:

- We never know the true function.
- We usually have only one sample dataset.
- Therefore, we cannot calculate the true Bias and Variance directly.

Instead, we **infer** them using practical tools:

Practical Tool	What it reveals
Training & Validation Performance	<u>Diagnoses basic overfitting vs underfitting</u>
Train–Validation–Test Split	<u>Verifies generalisation to truly unseen data</u>
Cross-Validation	<u>Checks model stability across data subsets</u>
Learning Curves	<u>Shows if model needs more data or capacity</u>

PRACTICE P4 · COMPREHENSIVE CHALLENGE*(Homework)*

(a) **Fill in the Key Terms table:**

Term	Definition
Estimator	<u>A rule/formula that estimates an unknown quantity using data.</u>
Bias	<u>The inability of a model to fit the sample training data.</u>
Variance	<u>How much the prediction changes when training data is changed.</u>
Underfitting	<u>Models too simple to capture patterns (high bias, low variance).</u>
Overfitting	<u>Models memorize specific details and noise (low bias, high variance).</u>
Irreducible Error	<u>Inherent noise in the data that no model can eliminate.</u>

(b) **True or False:**

Statement	T/F	Justification
The true Bias and Variance can always be calculated from a real-world dataset.	[F]	<u>We only have one sample, not infinite samples.</u>
$MSE = \text{Bias} + \text{Variance} + \text{Irreducible Error}$.	[F]	<u>It is Bias squared.</u>
A model with very high training accuracy but low validation accuracy likely has high variance.	[T]	<u>Definition of overfitting.</u>
Adding more data always fixes a high bias problem.	[F]	<u>It doesn't help if model lacks capacity.</u>
Cross-validation helps estimate the stability of model performance.	[T]	<u>It tests on multiple distinct splits.</u>

(c) **Find the Mistake.** A student says: “My model has 99% training accuracy and 98% validation accuracy. It must be suffering from high variance because the training accuracy is

very high.”

The mistake is looking only at training accuracy. Variance is indicated by a large gap (e.g.

- (d) **Predict the Outcome.** A data scientist uses a degree-1 polynomial to model a dataset where the true relationship is $y = x^3 + 2x^2 - x + 5$. What will happen to the Bias and Variance? Will the model underfit or overfit?

The model is a straight line, but the true data is a cubic curve. The model will suffer from

BY THE END OF THIS SHEET, YOU CAN

- ☐ Write the equation $Y = f(x) + \varepsilon$ and explain each term.
- ☐ Explain why $\hat{f}(x)$ is a random variable.
- ☐ Define Bias and Variance mathematically.
- ☐ State the Bias–Variance Decomposition: $\text{MSE} = \text{Bias}^2 + \text{Variance} + \text{Var}(\varepsilon)$.
- ☐ Use the dartboard analogy to visualise all four bias-variance combinations.
- ☐ Explain the Bias–Variance Tradeoff and its relationship to model complexity.
- ☐ Explain why true Bias and Variance cannot be computed in practice.
- ☐ Diagnose High Bias and High Variance from training/validation performance.
- ☐ Interpret learning curves to identify underfitting and overfitting.

Answer Key Summary

- P1** (a) Too simple: High Bias, Low Var, Underfits. Too complex: Low Bias, High Var, Overfits.
Approp: Low Bias, Low Var, Good fit.
(b) False. Underfitting has low variance; stable predictions.
(c) False. High bias means low variance.
- P2** (a) 2. (b) 8. (c) $1 \rightarrow B$, $2 \rightarrow C$, $3 \rightarrow D$, $4 \rightarrow A$. (d) False. (e) 12, Variance dominates.
- P3** (a) Good fit, High Bias, High Variance, High Bias.
(b) Overfitting. Limit depth, add data.
(c) High Variance, Yes, No.
(d) (i) Both true and R explains A.
- P4** (a) See definitions in text.
(b) F, F, T, F, T.
(c) Small gap = low variance, good fit.
(d) High bias, low variance, underfitting.